Product Design - Uber

Asked by : **Uber**

Problem Statement : How would you design an Uber-like car sharing platform for disabled users?

Clarifying Questions :

WHO (Target audience)

Q1 : Are we focusing on all disabled people or starting with a specific segment?

Ans : Focus on all types of disability.

Q2 : Do caregivers also book rides?

Ans **:** Yes — caregivers (family, NGOs, healthcare staff) are critical secondary users who often book on behalf of disabled passengers.

WHAT (Problem we're solving)

Q:3 Are we solving for accessible vehicles only, or end-to-end travel needs?

Ans **:** Both. Vehicles must be physically accessible, but the product must also solve for booking accessibility, communication barriers, and safety throughout the journey.

Q4: Will this be a new fleet or an extension of Uber's fleet?

Ans **:** To move fast, we'll retrofit and certify a subset of existing drivers/vehicles, while also exploring dedicated fleets in the long term.

WHY (Goal of this product)

Q5 : Why build this platform?

Ans **:** 1 in 7 people globally live with a disability, yet mobility options are very limited. This is both a **market gap** and a **social inclusion opportunity**. Beyond business, it aligns with regulatory pushes for inclusive transport.

WHEN (Usage patterns)

Q6: Is demand on-demand or scheduled?

Ans **:** Both. Many will need scheduled rides (medical appointments, work commutes), but on-demand rides are equally critical for daily independence.

WHERE (Geography)

Q7 : Which markets do we target first?

Ans **:** Start with urban metro cities (Delhi, Mumbai, Bangalore, NYC, London) where demand density supports dedicated accessible fleets. Expansion to tier-2 cities can follow.

User Identification:

- **Primary Customers**
    - People with physical disabilities (wheelchair users, limited mobility).
    - People with visual impairments (blind or low vision).
    - People with hearing/speech impairments.
    - People with cognitive disabilities (who may need simplified booking flows).
- **Secondary Customers**
    - Caregivers, family members who book rides on behalf of disabled users.

User Pain Points:

- **Physical Disabilities**
    - Lack of wheelchair-accessible vehicles.
    - Difficulty entering/exiting vehicles.
    - Inability to store wheelchairs safely.
- **Visual Disabilities**
    - Difficulty booking rides independently (app navigation issues).
    - Difficulty identifying the right car at pickup.
    - Lack of audio navigation support inside car.
- **Hearing / Speech Disabilities**
    - Communication barriers with drivers (directions, special needs).

  - No visual way to receive real-time alerts or driver instructions.
- **General Needs**
  - Safety & dignity (trained drivers, non-discriminatory treatment).
  - Reliability (on-time rides for hospital visits, jobs).
  - Affordability (rides should not be disproportionately expensive).
  - Assistance (help with door-to-door mobility).

Prioritize Pain Points:

Top Priorities

1. Availability of accessible vehicles (highest blocker).
2. Ease of booking (accessibility in app, caregiver booking).
3. Driver training & sensitivity (safety, dignity).
4. Assistive communication between passenger & driver (visual/audio/text modes).

Recommend Solutions:

- **Accessible Vehicle Fleet**
  - Retrofit cars with ramps and wheelchair locks.
  - Partner with OEMs & NGOs to build accessible fleets.
- **App Accessibility**
  - Voice-enabled booking (screen reader friendly).
  - Simplified interface with high-contrast UI and large fonts.
  - Caregiver booking & live tracking.
- **Driver Enablement**
  - Specialized training modules (assisting wheelchair users, basic sign language).
  - Badge/recognition for trained drivers.
  - Incentives for completing accessibility rides.
- **Assistive Communication**
  - In-app chat with quick pre-set messages for deaf/mute users.
  - Voice-to-text and text-to-voice communication tools.
  - Audio & vibration alerts for blind passengers.
- **Service Reliability**

- Advance booking option (for medical appointments).
- Integration with hospitals, care centers for bulk demand.

Prioritizing Solutions:

Solution	Reach (1–5)	Impact (1–5)	Confidence (%)	Effort (1–5)	RICE Score
Accessible App Experience (voice booking, caregiver mode, large-font UI)	5	4	100%	2	**10.0**
Driver Training & Certification (sensitivity training, incentives, badges)	4	5	100%	3	**6.7**
Accessible Vehicle Fleet (ramps, wheelchair locks, low-entry cars)	5	5	90%	5	**4.5**
Assistive Communication (text-to-speech, preset quick messages, vibration alerts)	3	4	80%	3	**3.2**
Reliability Layer (advance booking, hospital/NGO partnerships)	4	3	70%	4	**2.1**

Insights from Prioritization

- The **Accessible App** is the clear low-effort, high-impact starting point → MVP should ship this first.
- **Driver Training** is the next priority because even a few accessible vehicles fail without trained drivers.
- **Accessible Fleet** is critical but effort-heavy → must start in parallel but scale gradually.

- **Assistive Communication** and **Reliability Layer** are Phase 2/3 features.

Metrics:

- **Acquisition:** # of disabled users registering monthly.
- **Engagement:** Repeat rides per user, % of rides booked by caregivers.
- **Satisfaction:** NPS and CSAT scores specifically for disabled riders.
- **North Star Metric:** % increase in completed rides by disabled users per month